



Khulna University of Engineering & Technology

Department of Computer Science and Engineering

Project Report

QSync - Queue Management System

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Year : Second
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Course No : CSE 2200
Course Name : Advance Programming Laboratory
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(Desktop)

QSync - Queue Management System

Project Description

QSync is a comprehensive Desktop-based queue management system designed for those facilities to streamline appointment scheduling and queue management. The application provides a dual-interface solution with separate portals for patients and administrators, enabling efficient management of departments, doctors, and patient queues in real-time.

Key Features

- **Dual User Interface:** Separate portals for patients and administrators
- **Real-time Queue Management:** Live tracking of patient queues across departments
- **Database Integration:** Secure authentication and SQLite database synchronization.
- **Department & Doctor Management:** Admin tools to manage healthcare providers
- **Priority-based Queuing:** Support for normal, urgent, and emergency appointments
- **User Profiles:** Personalized user accounts with appointment history

Project Details

System Architecture

QSync Desktop is built on a modular architecture that emphasizes reliability and performance:

Core Technologies

- **Java:** Developed using Java 25, leveraging modern language features for performance and safety.
- **JavaFX:** Utilizes JavaFX 21 for the graphical user interface, providing a smooth and responsive desktop experience.
- **SQLite:** Uses SQLite 3 for local data persistence, ensuring fast access without the need for a separate database server.

Design Patterns

- **Model-View-Controller (MVC):** Decouples business logic from the user interface, allowing for independent development and testing of each component.
- **Data Access Object (DAO):** Provides an abstraction layer for database operations, ensuring that the application logic remains independent of the storage mechanism.
- **Service Layer:** Encapsulates core business processes (e.g., appointment scheduling, session management) to ensure code reusability and consistency.

Database Design

The application uses an embedded SQLite database (qsync.db) with a strictly enforced relational schema:

Authentication Tables:

- users: Stores patient profile information.
- admins: Stores administrative credentials.
- Security: All passwords are stored as cryptographic hashes (SHA-256).

Healthcare Hierarchy:

- departments: Categorizes medical services.
- doctors: Profiles medical staff with associations to specific departments.

Queue Management:

- appointments: The core table tracking patient tokens, schedules, and various statuses (Waiting, In Progress, Completed, Cancelled).
- Data Integrity: Implemented using foreign key constraints and triggered timestamps.

Authentication System

- User Login & Registration: Email/password authentication via Local Auth
- Admin Login: Separate admin authentication portal with role-based access control
- Auto-login: Automatic session management with token refresh
- Role Detection: RBAC (Role Bases Access Control) for admin and user login

User Portal

Implemented a complete patient-facing interface with **Sidebar** navigation:

- Home Fragment: Dashboard displaying overview of all data.
- Join Fragment:
 - Department and doctor selection
 - Priority level selection (Normal, Urgent, Emergency)
 - Symptoms/reason for visit input
 - Estimated wait time calculation
 - Queue position preview
- History Fragment: View past appointments and queue history
- Profile Fragment: User profile management and settings
- Sidebar: A sidebar for navigating to all pages and also a sign out button at the bottom.

Admin Portal

Comprehensive administrative dashboard with:

- Admin Home:
 - Live queue monitoring with expandable tree view
 - Real-time status updates (Waiting, In Progress, Completed, Cancelled)
 - Department and doctor-wise queue organization
- Manage Fragment: Central hub for administrative tasks
 - Department management (Add, Edit, Delete)
 - Doctor management (Add, Edit, Delete with department assignment)
- Admin Profile: Admin account settings and logout functionality
- Completed Fragment: Archive of completed appointments
- Sidebar: A sidebar for navigating to all pages and also a sign out button at the bottom.

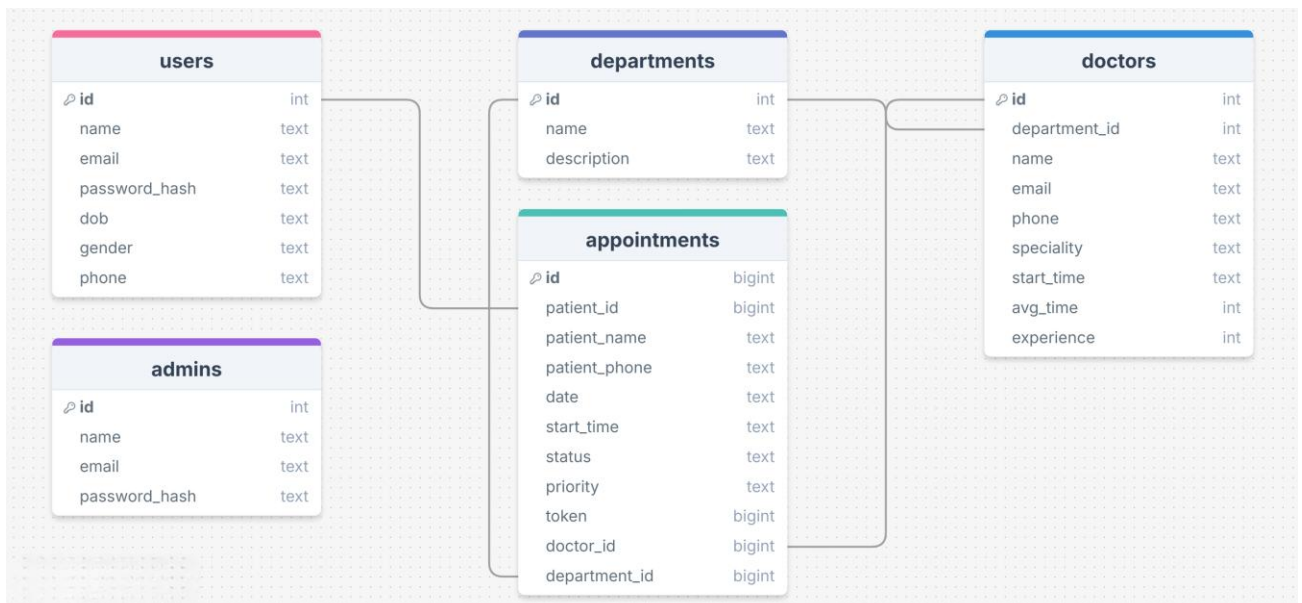
Data Models

- User: Patient information model
- Department: Healthcare department structure
- Doctor: Doctor profiles with department associations
- Queue: Appointment queue items with status tracking

Tech Stack

- Java
- JavaFX (Design)
- Maven (Build)
- SQLite (Database)

Database Schema



QSync - Login

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Join QSync

Create your account to start managing appointments.

Full Name

Enter name

Email

Enter email

Date of Birth

Gender

Male

Phone Number

Enter phone

Password

Create Account

Already have an account? [Sign in](#)

User Registration

QSync - Login

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QSync

Welcome back! Please enter your details.

Email address

Enter your email

Password

Sign in

Don't have an account? [Sign up](#)

User Login

QSync - Login

QSync

Dashboard

Medical history

Book Appointment

Settings

Sign out

Welcome back

mutiur5bb@gmail.com

Refresh

Book Appointment

Today's Appointments

Dr. Tanvir Hossain

Scheduled at: 06:30

Token: 2

waiting

urgent

Cancel

Dr. Tanvir Hossain

Scheduled at: 06:38

Token: 3

waiting

urgent

Cancel

Upcoming Appointments

Date	Doctor	Token	Status	Priority	Actions
2026-01-13	Dr. Ayesha Rahman	1	Waiting	urgent	Cancel
2026-01-13	Dr. Ayesha Rahman	2	Waiting	urgent	Cancel
2026-01-13	Dr. Farhana Akter	1	Waiting	normal	Cancel

User Dashboard

Book an Appointment

Choose your preferred doctor and schedule

Department

Neurology

Doctor

Dr. Ayesha Rahman

Preferred Date

1/13/2026

Symptoms / Notes

Briefly describe your symptoms

Priority

Normal

Urgent

Emergency

Confirm and Book Now

Message

Message

Booked appointment id: 7 (token: 2)

OK

Dr. Tanvir Hossain

Scheduled at: 06:38

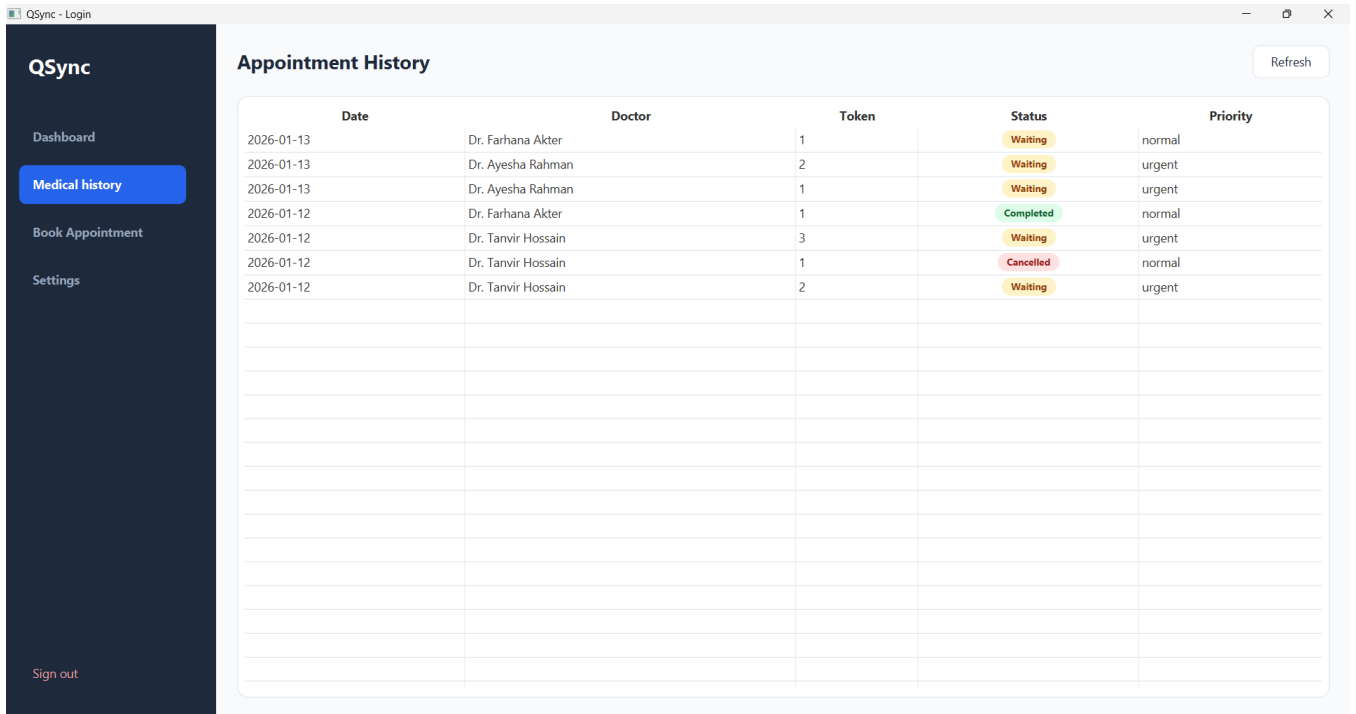
Token: 3

waiting

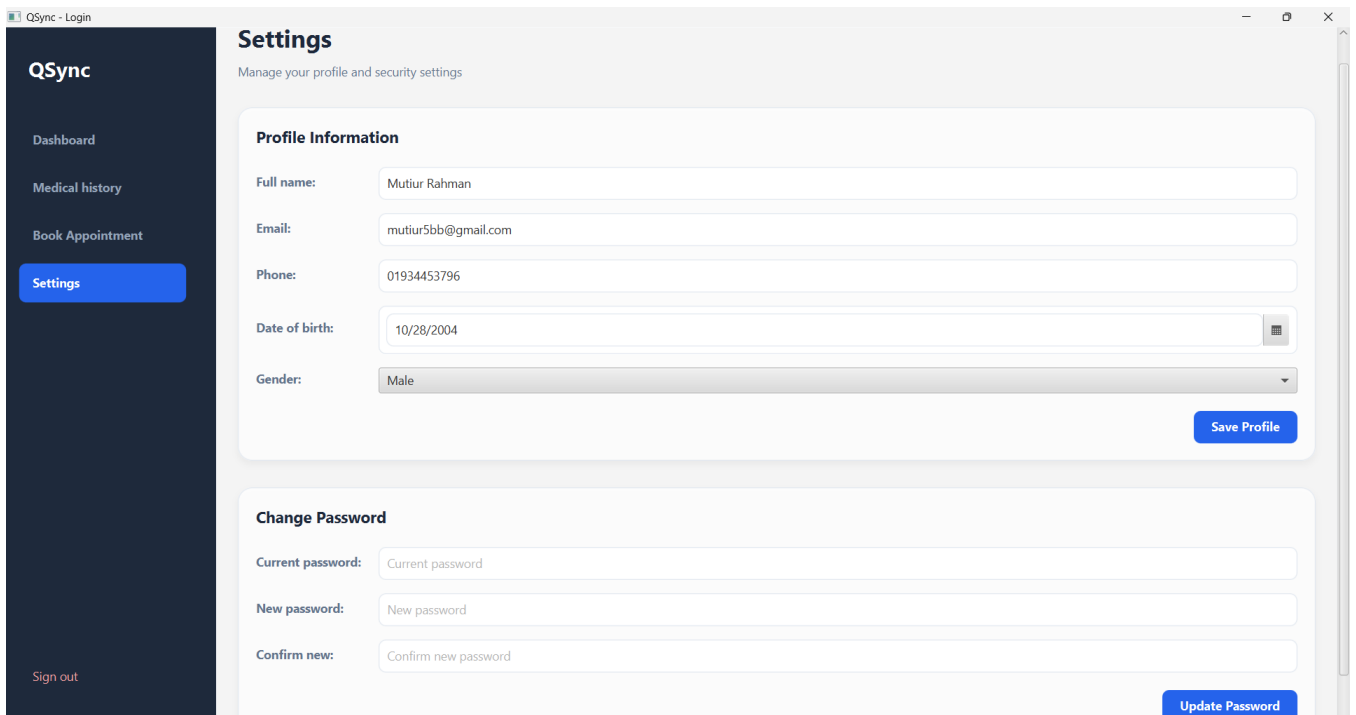
urgent

Cancel

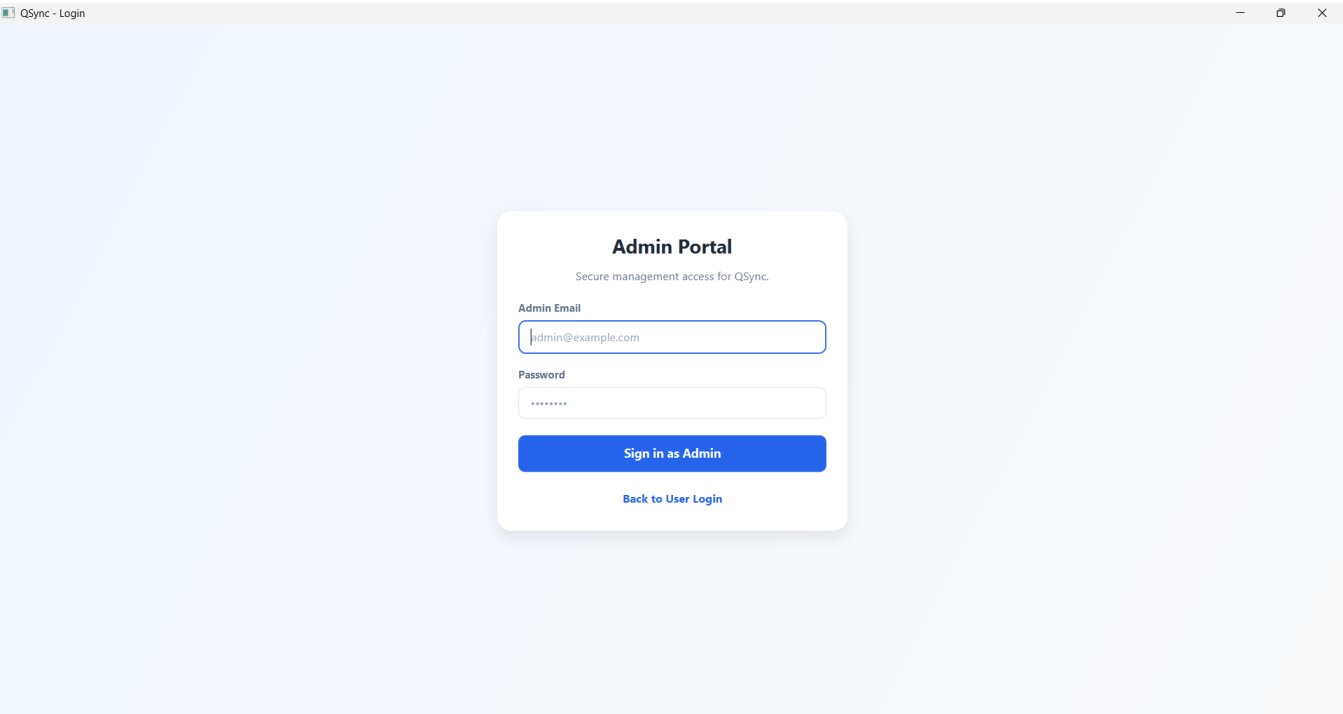
Queue Joining



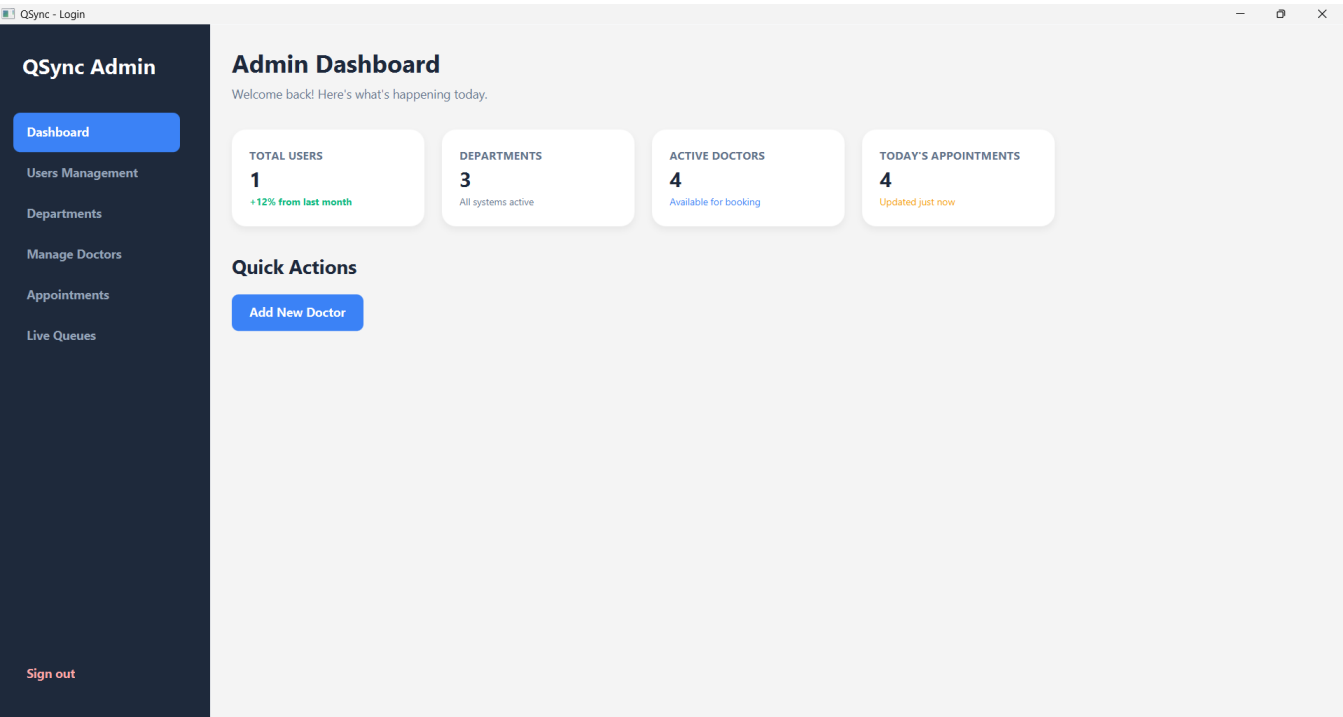
Appointment History



Profile



Admin Login



Admin Dashboard

QSync - Login

QSync Admin

Dashboard

Users Management

Departments

Manage Doctors

Appointments

Live Queues

Sign out

Live Patient Queues

Monitor real-time patient status by department and doctor for today.

Refresh

▼ Cardiology

▼ Dr. Dr. Farhana Akter

#1	Mutiur Rahman	09:00	normal	waiting
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▼ Neurology

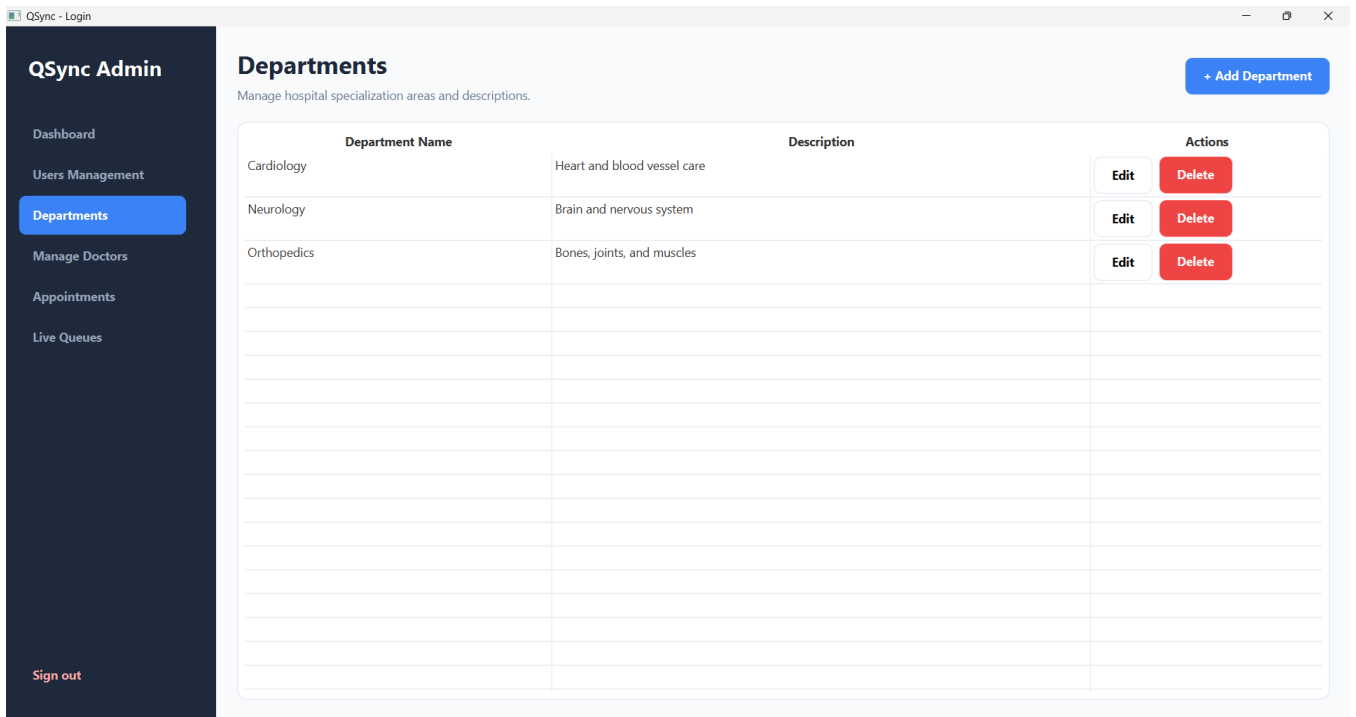
▼ Dr. Dr. Ayesha Rahman

#1	Mutiur Rahman	03:30	urgent	waiting
#2	Mutiur Rahman	03:42	urgent	waiting

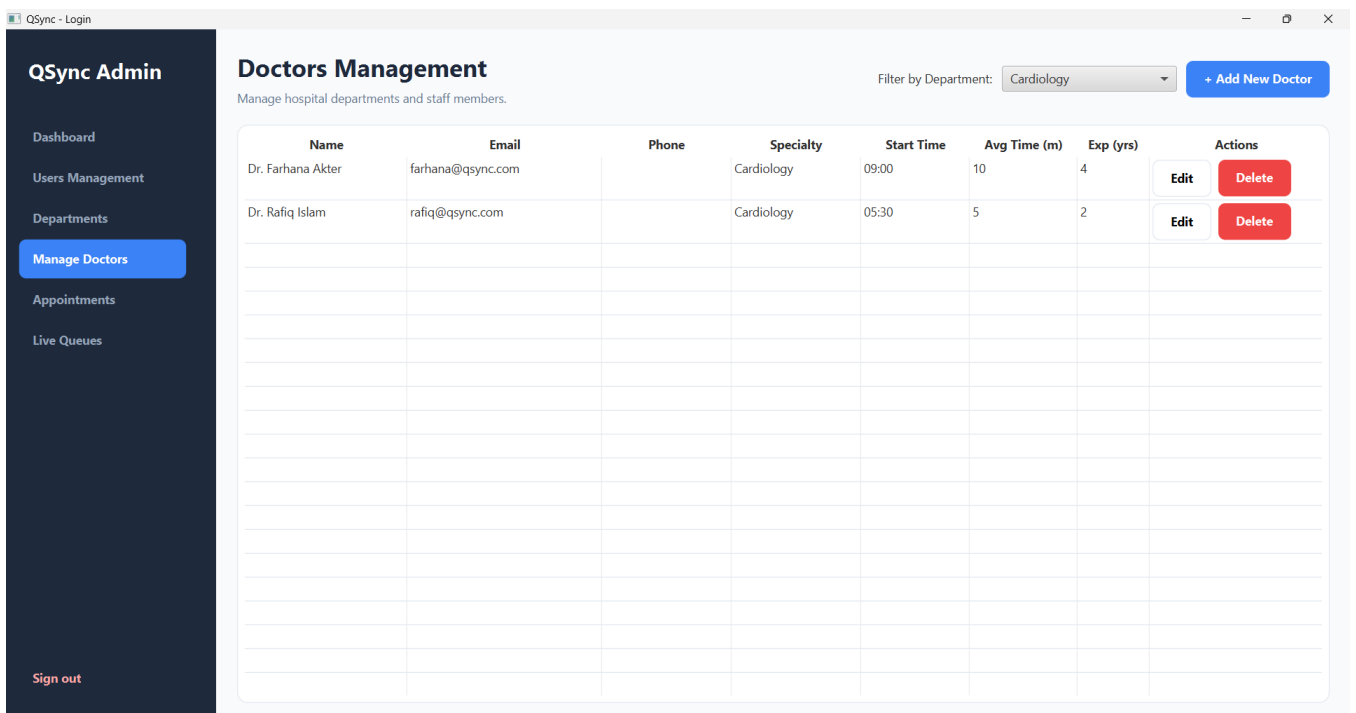
Today's Queues

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Appointment History



Department Management



Doctor Management